

Lovish Thukral

Software Engineer

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SUMMARY

Computer Science Engineering student with strong expertise in full-stack and systems-oriented development, specializing in the MERN stack and React Native. Experienced in building scalable backend systems, RESTful APIs, and integrating AI models into real-world applications. Possess solid understanding of modern software architecture, performance optimization, and asynchronous programming. Passionate about intelligent systems, robotics, and developing efficient, high-performance software solutions.

PROFESSIONAL EXPERIENCE

Vidya Corporation

06/2025 – 07/2025 | Ludhiana

Full Stack Developer Trainee

- Developed a full-stack social media application using the MERN stack with user authentication, posts, likes, and comments.
- Built RESTful APIs and managed MongoDB database for efficient data handling.
- Created responsive frontend using React and implemented secure authentication using JWT.
- Assisted in debugging, testing, and improving overall application performance.

EDUCATION

Bachelor of Technology (B.Tech) - Computer Science Engineering

07/2023 – Present

RIET, Phagwara

SKILLS

Frontend Development: — React.js, React Native, Tailwind, Bootstrap

Backend Development: — Node.js, Express.js, REST API Development, MongoDB, Supabase, Authentication (JWT), Database Design

AI & Systems: — LLM Integration, Prompt Engineering, Basic Transformer Architecture, LLAMA Models, AI Orchestration, ONNX Runtime, Basic Quantization and Fine-tuning

Dev Tools & Build Systems: — Git, GitHub, Postman, Linux, Docker

PROJECTS

Sabrina AI - Native AI Agent

AI Voice Assistant | Local LLM | Real-Time Command Execution

- Built SabrinaAI, a fully offline voice AI assistant with a push-to-listen STT → LLM → TTS pipeline.
- Triggered via shortcut key to start a session, capturing audio through ALSA with VAD (ONNX Runtime) for speech boundary detection and faster-whisper for transcription, feeding into llama.cpp for on-device LLM inference with a TTS response layer. No cloud dependencies, designed for low-latency edge deployment on consumer hardware.

Nextep AI – Intelligent Career Mentorship & Interview Platform

React.js | Node.js | MongoDB | Express | Groq API | Tailwind CSS | Axios

- Developed an intelligent backend leveraging MongoDB for flexible data persistence and Groq API for high-speed inference, utilizing advanced prompt engineering to optimize LLM responses.
- Architected a responsive frontend using modern frameworks to provide a seamless user interface for interacting with AI agents and visualizing complex data.

Bliss

Full Stack Social Platform | MERN Stack | REST API Integration

- Developed a full-stack social media application using the MERN stack.
- Implemented user authentication, posts, likes, and comments with secure REST APIs.
- Built responsive frontend using React Native and managed data with MongoDB.
- Designed scalable backend architecture for efficient data handling.

ACHIEVEMENTS

Participant - Hack-n-Win Hackathon

Engineered an end-to-end career development platform featuring a voice-integrated mock interview system using React, Node.js, and MongoDB.

Open Source Contributor | GirlScript Summer of Code (GSSOC)

Contributed to multiple high-impact repositories by optimizing codebase performance and implementing new features across diverse tech stacks.

INTERESTS

Robotics & Intelligence Systems | Emerging Technologies